1. Experiments and Results

4a. Varietal Classification

The classification of wheat seeds into 50 distinct varieties, referred to as *varietal classification*, was performed using both 2D and 3D convolutional neural networks. The results are summarized in Table 5. The "Epochs" column reports the number of training epochs completed before early stopping, with a maximum limit of 100. The 3D-ResNet model converged in approximately 50 epochs, while the other 3D models typically required around 80 epochs to complete training. All 3D models achieved a training accuracy of 0.99. Among them, the 3D-ResNet yielded the highest test accuracy (0.98), followed by 3D-DenseNet (0.96) and 3D-ConvNeXt (0.94). The average precision, recall, F1-score, and kappa coefficient exhibited a similar trend. These results demonstrate that the 3D-ResNet model is the most effective architecture for bulk wheat seed classification.

Overall, the 2D models exhibited lower performance compared to their 3D counterparts, with DenseNet and ConvNeXt following this general trend. Notably, the 2D DenseNet model underperformed, achieving results below the average. However, both the 3D and 2D versions of ResNet delivered nearly equivalent classification performance. The primary distinction between them lies in training efficiency — 3D-ResNet trained significantly faster than its 2D counterpart. Further analysis of memory consumption and training time per epoch indicated that although 2D models consume slightly less memory, training speed emerged as a more critical factor. In this regard, 3D models demonstrated clear advantages, with 3D-ResNet being the fastest among all evaluated architectures.

Model	Epochs	Training Accuracy	Validation Accuracy	Test Accuracy	Avg. Precision	Avg. Recall	Avg. F1 Score	Kappa Coefficient	Memory (MiB)	Train Time (min)
3D-ResNet	54/100	0.9996	0.9666	0.9845	0.9878	0.9845	0.9844	0.9842	15875	162
3D-DenseNet	87/100	0.9952	0.9358	0.9635	0.9665	0.9635	0.962	0.9628	15227	319
3D-ConvNext	84/100	1.0000	0.9541	0.9365	0.9398	0.9364	0.9331	0.9352	12277	378
2D-ResNet	55/100	0.9999	0.9934	0.9951	1.0000	1.0000	1.0000	0.9950	14889	431
2D-DenseNet-121	45/100	0.9832	0.1101	0.1176	-	-	-	0.0996	11313	368
2D-ConvNext-T	27/100	0.9714	0.7186	0.6972	0.8278	0.8118	0.7920	0.6910	12365	225

Table 5: The results of 2D and 3D CNN models for varietal classification of wheat seeds

Next, the learning curves for the 3D-ResNet, 3D-DenseNet, and 3D-ConvNeXt models are analyzed and shown in Figure 4. The 3D-ResNet model exhibits large fluctuations during the first 25 epochs before stabilizing for the remainder of the training. The training is completed in 54 epochs. Similarly, the 3D-DenseNet model experiences fluctuations for approximately half of the training process before stabilizing and finally converging at 87 epochs. In contrast, the 3D-ConvNeXt model demonstrates the most stable learning curves throughout the entire training process.

Figure 5 presents the confusion matrices for the 3D-ResNet, 3D-DenseNet, and 3D-ConvNeXt models. The 3D-ResNet model shows significant misclassifications for the RAJ1482_R_22 and RAJ3077_R_22 species. The 3D-DenseNet model exhibits similar deviations, additionally misclassifying RAJ4079_R_22. In contrast, the 3D-ConvNeXt model demonstrates misclassifications for RAJ4083_R_22, DDK1029_21_22_K, and MACS2971_21_22_K species. These results suggest that while 3D-ResNet and 3D-DenseNet exhibit similar learning patterns, the 3D-ConvNeXt model follows a slightly different learning behavior.

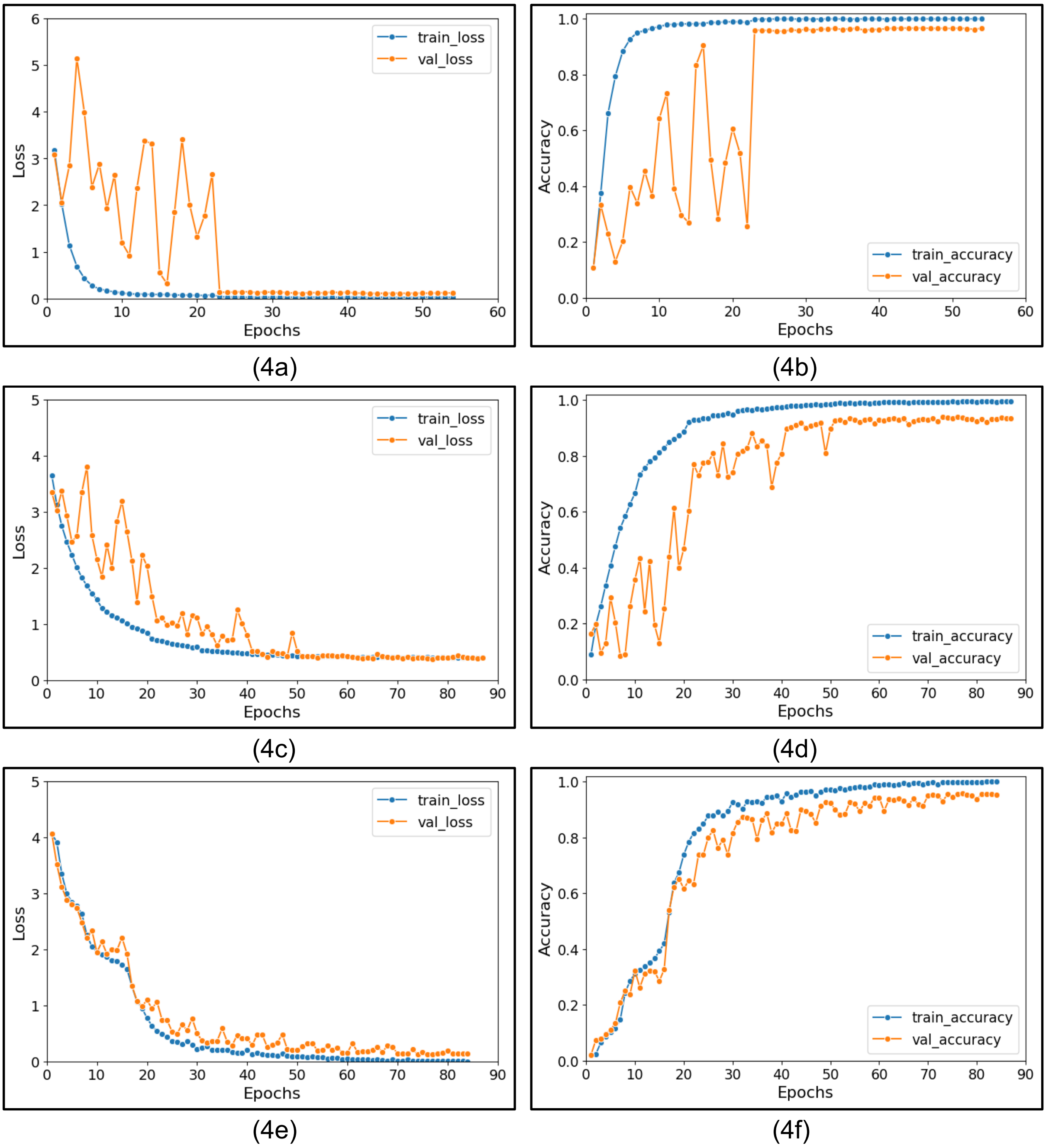
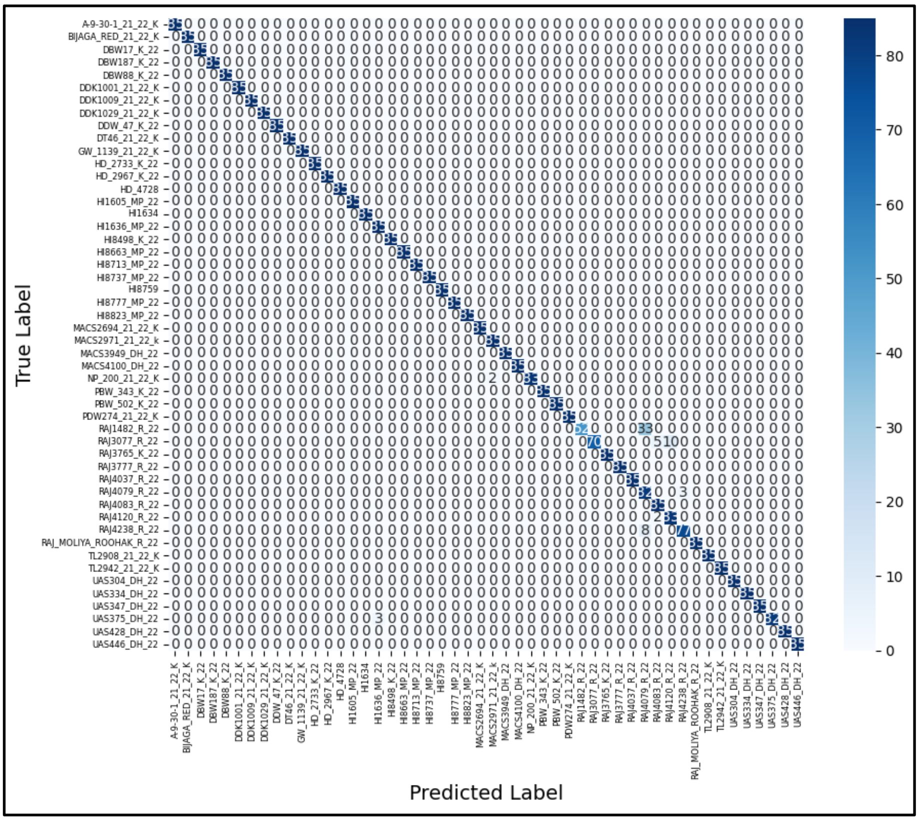
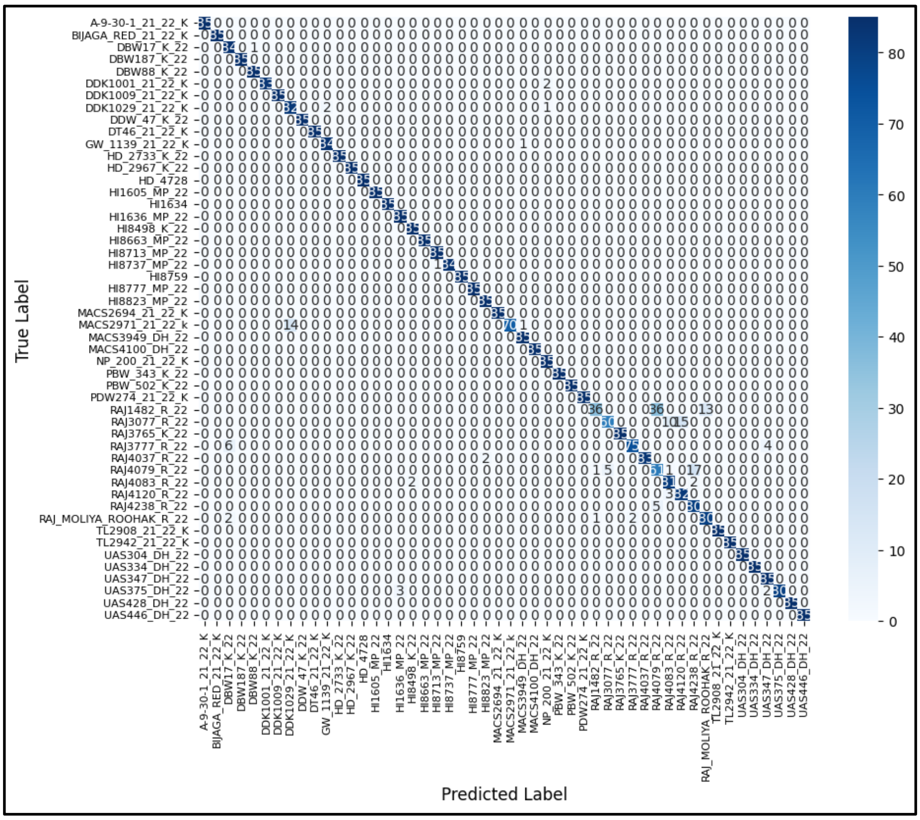


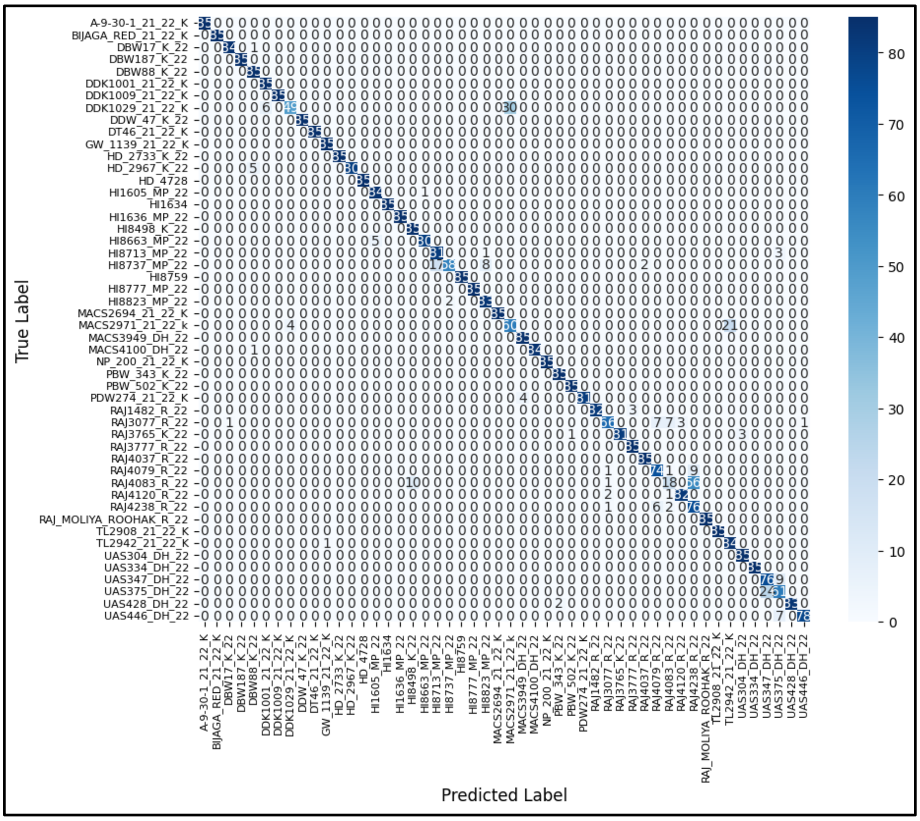
Figure 4: The plots in (4a) and (4b) represent the loss and accuracy curves for the 3D-ResNet model, similarly (4c) and (4d) correspond to the 3D-DenseNet-121 model, and (4e) and (4f) illustrate the results for the 3D-ConvNeXt-T model


True Label
Predicted Label
A-9-30-1_21_22_K
BIJAGA_RED_21_22_K
DBW17_K_22
DBW187_K_22
DBW88_K_22
DDK1001_21_22_K
DDK1009_21_22_K
DDK1029_21_22_K
DDW_47_K_22
DT46_21_22_K
GW_1139_21_22_K
HD_2733_K_22
HD_2967_K_22
HD_4728
HI1605_MP_22
HI1634
HI1636_MP_22
HI8498_K_22
HI8663_MP_22
HI8713_MP_22
HI8737_MP_22
HI8759
HI8777_MP_22
HI8823_MP_22
MACS2694_21_22_K
MACS2971_21_22_k
MACS3949_DH_22
MACS4100_DH_22
NP_200_21_22_K
PBW_343_K_22
PBW_502_K_22
PDW274_21_22_K
RAJ1482_R_22
RAJ3077_R_22
RAJ3765_K_22
RAJ3777_R_22
RAJ4037_R_22
RAJ4079_R_22
RAJ4083_R_22
RAJ4120_R_22
RAJ4238_R_22
RAJ_MOLIYA_ROOHAK_R_22
TL2908_21_22_K
TL2942_21_22_K
UAS304_DH_22
UAS334_DH_22
UAS347_DH_22
UAS375_DH_22
UAS428_DH_22
UAS446_DH_22
80
70
60
50
40
30
20
10
0

(5a)


True Label
Predicted Label
A-9-30-1_21_22_K
BIJAGA_RED_21_22_K
DBW17_K_22
DBW187_K_22
DBW88_K_22
DDK1001_21_22_K
DDK1009_21_22_K
DDK1029_21_22_K
DDW_47_K_22
DT46_21_22_K
GW_1139_21_22_K
HD_2733_K_22
HD_2967_K_22
HD_4728
HI1605_MP_22
HI1634
HI1636_MP_22
HI8498_K_22
HI8663_MP_22
HI8713_MP_22
HI8737_MP_22
HI8759
HI8777_MP_22
HI8823_MP_22
MACS2694_21_22_K
MACS2971_21_22_k
MACS3949_DH_22
MACS4100_DH_22
NP_200_21_22_K
PBW_343_K_22
PBW_502_K_22
PDW274_21_22_K
RAJ1482_R_22
RAJ3077_R_22
RAJ3765_K_22
RAJ3777_R_22
RAJ4037_R_22
RAJ4079_R_22
RAJ4083_R_22
RAJ4120_R_22
RAJ4238_R_22
RAJ_MOLIYA_ROOHAK_R_22
TL2908_21_22_K
TL2942_21_22_K
UAS304_DH_22
UAS334_DH_22
UAS347_DH_22
UAS375_DH_22
UAS428_DH_22
UAS446_DH_22
80
70
60
50
40
30
20
10
0

(5b)



(5c)

Figure 5: Confusion Matrices for 3D-ResNet, 3D-DenseNet and 3D-ConvNext Models in figures (5a), (5b) and (5c) respectively